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Tables and Figures

Table 1: True hazard ratios.

Covariate	**Type**	**Hazard ratio**
X_1	Binary	0.80
X_2	Continuous	1.05
X_3	Continuous	1.25
X_4	Binary	1.54
X_5	Binary	1.18
X_6	Continuous	1.45
X_7	Binary	1.10
X_8	Binary	0.76
X_9	Binary	0.64
X_{10}	Binary	1.25
X_{11}	Binary	0.48

Table 2: Covariance matrix for simulated data.

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}
X_1	0.2370										
X_2	-1.3349	196.2990									
X_3	0.0812	0.6471	1.0967								
X_4	0.0247	-0.6314	0.0680	0.2441							
X_5	0.0298	-0.0759	0.0452	0.0063	0.1302						
X_6	0.0134	0.3167	0.0063	-0.0089	0.0070	0.0558					
X_7	0.0280	-0.7944	0.0740	0.0468	0.0214	-0.0015	0.2427				
X_8	-0.0069	0.0156	-0.0190	-0.0581	-0.0026	0.0050	-0.0119	0.1146			
X_9	0.0039	-0.1261	-0.0080	0.0484	0.0016	-0.0026	0.0131	-0.0318	0.1408		
X_{10}	0.0002	0.0294	-0.0147	0.0003	-0.0001	0.0003	0.0017	0.0001	0.0050	0.1369	
X_{11}	0.0223	-0.8057	-0.0075	-0.0139	0.0043	-0.0000	0.0012	0.0086	-0.0267	0.0014	0.0986

Table 3: Fixed probability of missingness parameters.

Missing covariate	k_j	l_j	γ_{2j}
X_1	—	2	$\log(1.05)$
X_3	1	5	$\log(0.80)$
X_4	3	6	$\log(0.70)$
X_7	4	8	$\log(0.90)$
X_{10}	5	9	$\log(0.60)$
X_{11}	10	2	$\log(1.05)$

Table 4: Marginal and joint missing data proportions.

Marginal missingness X_i	Marginal missingness X_j	Joint missingness (X_i, X_j)
0.05	0.05	0.01
0.05	0.15	0.02
0.15	0.15	0.05
0.15	0.30	0.07
0.30	0.30	0.10
0.30	0.60	0.20
0.60	0.60	0.40

Figure 1: Distribution of bias (y-axis) of AUC estimators (panels) comparing the CC approach and the BI approach with three imputation methods (color) when the sample size is 750 across missing data patterns (x-axis).

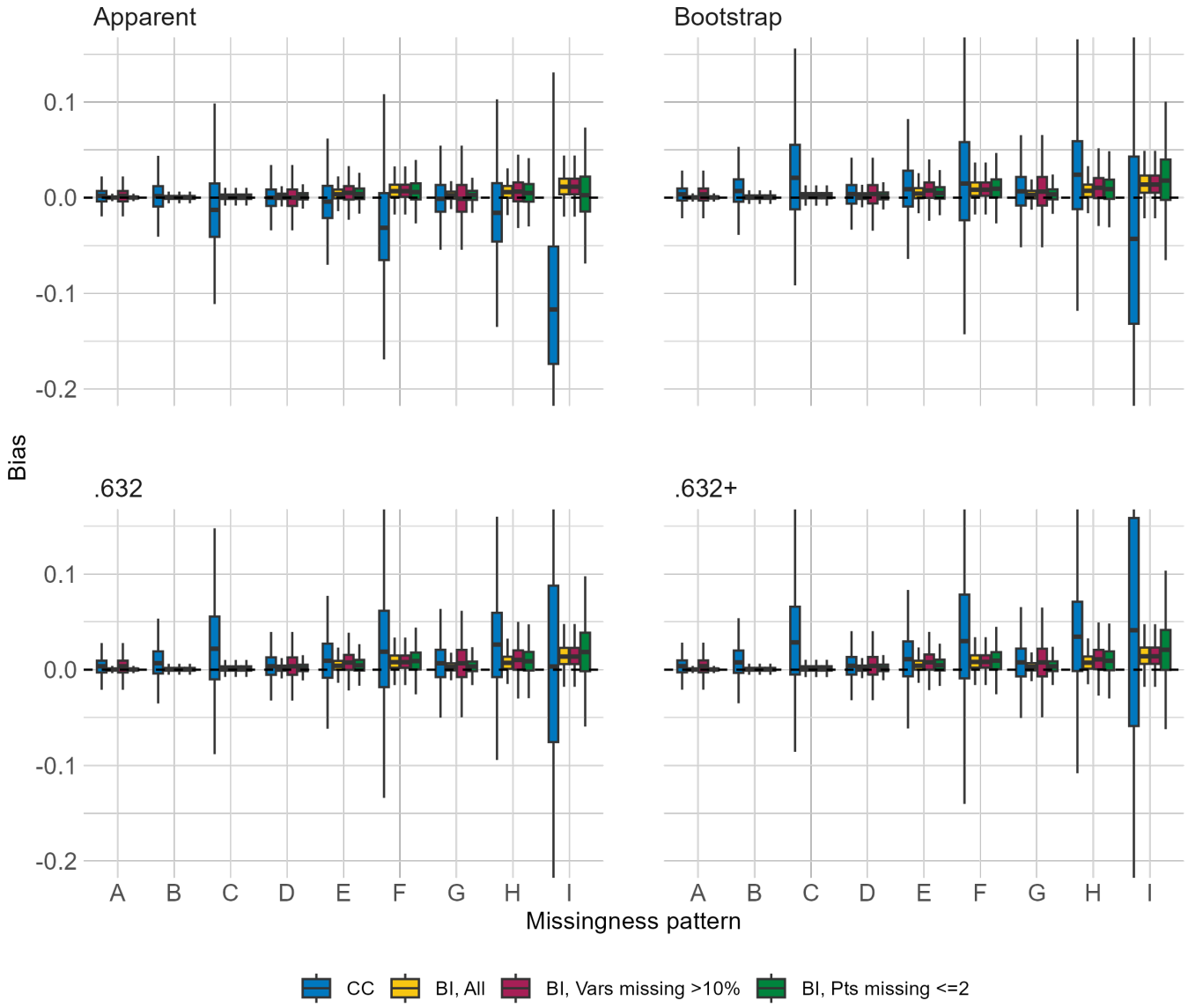
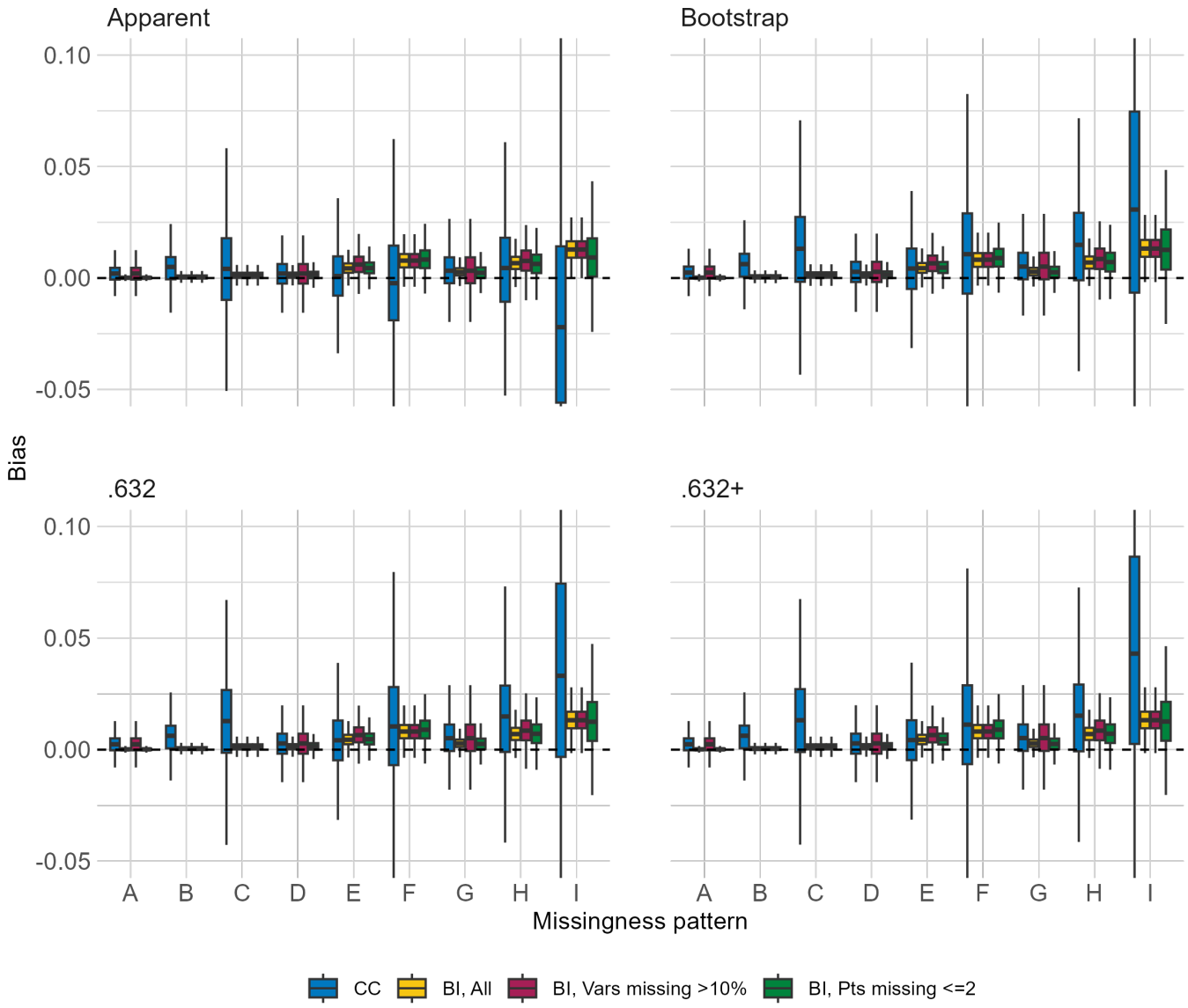
Figure 2: Distribution of bias (y-axis) of AUC estimators (panels) comparing the CC approach and the BI approach with three imputation methods (color) when the sample size is 3500 across missing data patterns (x-axis).